

3. CALCULUS

Rolle's theorem:

Let $f(x)$ is a function defined in $[a, b]$

$\Rightarrow$ a) $f(x)$ is continuous in $[a, b]$

b) $f(x)$ is differentiable (a, b)

$$f(a) = f(b)$$

Then there exist $c \in (a, b) \Rightarrow f'(c) = 0$

Verify Rolle's theorem for $f(x) = e^x [\sin x - \cos x]$ in $[\frac{\pi}{4}, \frac{5\pi}{4}]$

since $e^x, \sin x, \cos x$ are continuous functions $f(x)$ is continuous.

$$f'(x) = e^x [\cos x + \sin x] + (\sin x - \cos x) e^x$$

$f(x)$ is differentiable.

$$f\left(\frac{\pi}{4}\right) = e^{\pi/4} \left[\sin \frac{\pi}{4} - \cos \frac{\pi}{4}\right]$$

$$= e^{\pi/4} \left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right] = 0$$

$$f\left(\frac{5\pi}{4}\right) = e^{5\pi/4} \left[\sin \frac{5\pi}{4} - \cos \frac{5\pi}{4}\right] = e^{5\pi/4} \left[-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right] = 0$$

$$f(a) = f(b)$$

As all the conditions of Rolle's theorem are satisfied there

$$\text{exist } c \in \left[\frac{\pi}{4}, \frac{5\pi}{4}\right] \Rightarrow f'(c) = 0$$

$$\Rightarrow f'(c) = e^c [\cos c + \sin c] + e^c [\sin c - \cos c] = 0$$

$$e^c \sin c = 0$$

$$c = 0, \pi$$

$$c = \pi \in \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$$

$$f(x) = x^3 - 6x^2 + 11x - 6 \text{ in } [1, 3]$$

since every polynomial function is continuous

a) $f(x)$ is continuous.

$$f'(x) = 3x^2 - 12x + 11$$

$\Rightarrow f(x)$ is differentiable in $(1, 3)$

$$f(3) = (3)^3 - 6(3)^2 + 11(3) - 6 = 27 - 54 + 33 - 6 = 0$$

$$f(1) = 1 - 6(1) + 11(1) - 6$$

$$= 1 - 6 + 11 - 6 = 0$$

since all the conditions of Rolle's theorem is satisfied there

exist $c \in [1, 3] \Rightarrow f'(c) = 0$

$$f'(c) = 3c^2 - 12c + 11 = 0$$

$$c = \frac{12 \pm \sqrt{144 - 132}}{6}$$

$$= \frac{12 \pm \sqrt{12}}{6}$$

$$= \frac{12 \pm 2\sqrt{3}}{6}$$

$$= \frac{6 \pm \sqrt{3}}{3}$$

$$c = 2.57, 1.4$$

$\therefore$ Rolle's theorem is verified.

Q) $f(x) = \frac{1}{x^2}$ in $[-1, 1]$

sol) since $f(x) = \frac{1}{x^2} = \infty$ at $x=0$

$\Rightarrow f(x)$ is discontinuous at $x=0$

Rolle's theorem is not applicable.

Q) $f(x) = x^2$ $[1, 3]$

a) $f(x)$ is continuous.

b) $f'(x) = 2x$

$f(x)$ is differentiable $(1, 3)$

c) $f(1) = 1$

$f(3) = 9$

$f(1) \neq f(3)$

$\therefore$ Rolle's theorem is not applicable.

a) $f(x) = (x-a)^m (x-b)^n$ $[a, b]$

sol) a) $f(x)$ is continuous in $[a, b]$

b) $f'(x) = (x-a)^m n (x-b)^{n-1} + (x-b)^n m (x-a)^{m-1}$

$f(x)$ is differentiable

c) $f(a) = 0 = f(b)$

As all conditions of Rolle's theorem satisfied

$\exists c \in (a, b) \ni f'(c) = 0$

$$m(c-a)^{m-1} (c-b)^n + n(c-a)^m (c-b)^{n-1} = 0$$

$$(c-a)^{m-1} (c-b)^{n-1} \cdot (m(c-b) + n(c-a)) = 0$$

$$m(c-b) + n(c-a) = 0$$

$$mc - mb + nc - na = 0$$

$$c(m+n) = mb + na$$

$$c = \frac{mb + na}{m+n} \in (a, b)$$

a) verify Rolle's theorem for $x = \log \left[\frac{x^2 + ab}{x(a+b)} \right] [a, b]; a > 0, b > 0$

sol) a) $f(x)$ is continuous in $[a, b]$

b) $f(x) = \log(x^2 + ab) - \log x(a+b)$

$$= \log(x^2 + ab) - [\log x + \log(a+b)]$$

$$= \log(x^2 + ab) - \log x - \log(a+b)$$

$$f'(x) = \frac{1}{x^2 + ab} (2x) - \frac{1}{x} - 0$$

$$= \frac{2x}{x^2 + ab} - \frac{1}{x}$$

$f(x)$ is differentiable in (a, b)

c) $f(a) = \log \left[\frac{a^2 + ab}{a^2 + ab} \right] = \log 1 = 0$

$$f(b) = \log \left[\frac{b^2 + ab}{ba + b^2} \right] = \log 1 = 0$$

$$f(a) = f(b)$$

Since all conditions of Rolle's theorem are satisfied $\exists c \in (a, b)$

$$f'(c) = 0$$

$$\frac{2c}{c^2 + ab} - \frac{1}{c} = 0$$

$$\frac{2c^2 - c^2 - ab}{c(c^2 + ab)} = 0$$

$$c^2 - ab = 0$$

$$c = \sqrt{ab} \in (a, b)$$

Q) verify Rolle's theorem for $f(x) = x(x+3)e^{-x/2}$ $[-3, 0]$

Sol) a) $f(x)$ is continuous in $[-3, 0]$

$$\begin{aligned} b) f'(x) &= (x+3)e^{-x/2} + x e^{-x/2} (-1) + x(x+3) \left[-e^{-x/2} \cdot \frac{1}{2} \right] \\ &= (x+3)e^{-x/2} + x e^{-x/2} - \frac{1}{2} e^{-x/2} \cdot x(x+3) \end{aligned}$$

$f(x)$ is differentiable

$$\hookrightarrow f(-3) = 0$$

$$f(0) = 0$$

$$f(a) = f(b)$$

As all the conditions of Rolle's theorem are satisfied $\exists c \in (a, b)$

$$\exists f'(c) = 0$$

$$f'(c) = 0$$

$$(c+3)e^{-c/2} + c e^{-c/2} - \frac{1}{2} c(c+3)e^{-c/2} = 0$$

$$e^{-c/2} \left[c+3+c - \frac{1}{2}c^2 - \frac{3}{2}c \right] = 0$$

$$2c - \frac{1}{2}c^2 - \frac{3}{2}c + 3 = 0$$

$$-\frac{1}{2}c^2 + \frac{1}{2}c + 3 = 0$$

$$c^2 - 3c + 2c - 6 = 0$$

$$c(c-3) + 2(c-3) = 0$$

$$c \in (-2, 3)$$

$$c = -2 \in [-3, 0]$$

Q1) $f(x) = \sin x \sqrt{\cos 2x} \quad [0, \frac{\pi}{4}]$

sol) a) $f(x)$ is continuous in $[0, \frac{\pi}{4}]$

$$b) f'(x) = \sin x \cdot d(\sqrt{\cos 2x}) + \sqrt{\cos 2x} \cdot d(\sin x)$$

$$= \sin x \cdot \frac{1}{2\sqrt{\cos 2x}} (-\sin 2x) \cdot 2 + \sqrt{\cos 2x} \cdot \cos x$$

$$= -\frac{\sin x \cdot \sin 2x}{\sqrt{\cos 2x}} + \cos x \cdot \sqrt{\cos 2x}$$

$f(x)$ is differentiable in $(0, \frac{\pi}{4})$

$$c) f(a) = \sin 0 \sqrt{\cos 2(0)} = 0$$

$$f\left(\frac{\pi}{4}\right) = \sin \frac{\pi}{4} \sqrt{\cos 2 \cdot \frac{\pi}{4}} = 0$$

$$f(a) = f(c)$$

As all conditions of Rolle's theorem are satisfied $\exists c \in (0, \frac{\pi}{4})$ s.t.

$$f'(c) = -\frac{\sin c \sin 2c}{\sqrt{\cos 2c}} + \cos c \cdot \sqrt{\cos 2c} = 0$$

$$-\sin c \sin 2c + \cos c \cos 2c = 0$$

$$\cos(c+2c) = 0$$

$$\cos 3c = 0$$

$$\cos 3c = \cos \frac{\pi}{2}$$

$$3c = \frac{\pi}{2}$$

$$c = \frac{\pi}{6} \in (0, \frac{\pi}{4}]$$

Lagrange's mean value theorem:

→ let $f(x)$ be a function defined in $[a, b]$ &

i) $f(x)$ is continuous in $[a, b]$

ii) $f(x)$ is differentiable in (a, b)

$$\text{then } \exists c \in (a, b) \text{ s.t. } f'(c) = \frac{f(b) - f(a)}{b - a}$$

Q) verify Lagrange's mean value theorem $f(x) = x^3 - x^2 - 5x + 3$ $[0, 4]$

Sol) ① $f(x)$ is continuous in $[0, 4]$

$$\text{② } f'(x) = 3x^2 - 2x - 5$$

$f(x)$ is differentiable in $(0, 4)$

$$\therefore \exists c \in (0, 4) \text{ s.t. } f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f(4) = 64 - 16 - 20 + 3$$

$$f(0) = 3$$

$$f'(c) = \frac{31 - 3}{4 - 0} = \frac{28}{4} = 7$$

$$3c^2 - 2c - 5 = 7$$

$$3c^2 - 2c - 12 = 0$$

$$= \frac{2 \pm \sqrt{4 - 4(3)(-12)}}{6}$$

$$= \frac{2 \pm \sqrt{160}}{6}$$

$$= \frac{1 \pm \sqrt{39}}{3}$$

$$c = 2.36, -1.69,$$

Q) $f(x) = \log_e x$ in $[1, e]$

sol) a) $f(x) = \log_e x$ is continuous in $[1, e]$

b) $f'(x) = \frac{1}{x}$
 $f(x)$ is differentiable in $(1, e)$

$\therefore$ By LMVT $\exists c \in (1, e) \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$

$$\frac{1}{c} = \frac{f(e) - f(1)}{e - 1}$$

$$= \frac{\log_e e - \log_e 1}{e - 1} \quad [e' = 2.718]$$

$$\frac{1}{c} = \frac{1 - 0}{e - 1}$$

$$c = e' - 1 \in (1, e)$$

$$c = 2.718 - 1 = 1.718$$

Q) $f(x) = x(x-2)(x-3)$ in $[0, 4]$

sol) a) $f(x)$ is continuous in $[0, 4]$

b) $f'(x) = (x-2)(x-3) + x(x-3) + x(x-2)$
 $\therefore f(x)$ is differentiable in $(0, 4)$

By Lagrange's mean value theorem $\exists c \in (0, 4) \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$

$$(c-2)(c-3) + c(c-3) + c(c-2) = \frac{8 - 0}{4} = 2$$

$$c^2 - 3c - 2c + 6 + c^2 - 3c + c^2 - 2c = 2$$

$$3c^2 - 10c + 4 = 0$$

$$c = \frac{10 \pm \sqrt{100 - 4(3)(4)}}{2(3)} = \frac{10 \pm \sqrt{100 - 48}}{6}$$

$$= \frac{5 \pm \sqrt{13}}{3} = 0.46, 2.86 \in (0, 4)$$

Q) If $a < b$ p.t $\frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2}$ also deduce the following

$$\textcircled{a} \frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) < \frac{\pi}{4} + \frac{1}{6}$$

$$\textcircled{b} \frac{5\pi+4}{20} < \tan^{-1}2 < \frac{\pi+2}{4}$$

sol) let $f(x) = \tan^{-1}x$ defined in $[a, b]$ t.a.o $a > 0, b > 0$

a) $f(x)$ is continuous in $[a, b]$

$$\textcircled{b} f'(x) = \frac{1}{1+x^2}$$

$f(x)$ is differentiable in (a, b)

By LMVT $\exists c \in (a, b) \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$

$$\frac{1}{1+c^2} = \frac{\tan^{-1}(b) - \tan^{-1}(a)}{b - a} \rightarrow \textcircled{1}$$

$$c \in (a, b)$$

$$a < c < b$$

$$a^2 < c^2 < b^2$$

$$1+a^2 < 1+c^2 < 1+b^2$$

$$\frac{1}{1+a^2} < \frac{1}{1+c^2} < \frac{1}{1+b^2}$$

$$\frac{1}{1+b^2} < \frac{1}{1+c^2} < \frac{1}{1+a^2}$$

$$\frac{1}{1+b^2} < \frac{\tan^{-1}b - \tan^{-1}a}{b - a} < \frac{1}{1+a^2}$$

$$\frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2}$$

a) put $a=1; b=\frac{4}{3}$ ($\because \tan^{-1}(1) = \frac{\pi}{4}$)

$$\frac{\frac{4}{3}-1}{1+\frac{16}{9}} < \tan^{-1}\frac{4}{3} - \tan^{-1}1 < \frac{\frac{4}{3}-1}{1+1^2}$$

$$\frac{1}{3} \times \frac{9}{25} < \tan^{-1} \frac{4}{3} - \frac{\pi}{4} < \frac{1}{3} \times \frac{1}{2}$$

$$\frac{\pi}{4} + \frac{3}{25} < \tan^{-1} \frac{4}{3} < \frac{1}{6} + \frac{\pi}{4}$$

⑥ Put $b=2, a=1$

$$\frac{2-1}{1+4} < \tan^{-1} 2 - \tan^{-1} 1 < \frac{2-1}{1+1}$$

$$\frac{1}{5} < \tan^{-1} 2 - \frac{\pi}{4} < \frac{1}{2}$$

$$\frac{\pi}{4} + \frac{1}{5} < \tan^{-1} 2 < \frac{1}{2} + \frac{\pi}{4}$$

$$\frac{5\pi+4}{20} < \tan^{-1} 2 < \frac{\pi+2}{4}$$

Q) Prove that $\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < \sin^{-1} \frac{3}{5} < \frac{\pi}{6} + \frac{1}{8}$

Sol let $f(x) = \sin^{-1} x$ defined in $[a, b]$

a) $f(x)$ is continuous in $[a, b]$

$$b) f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$f(x)$ is differentiable in (a, b)

By LMVT $\exists c \in (a, b) \ni f'(c) = \frac{f(b) - f(a)}{b - a}$

$$\frac{1}{\sqrt{1-c^2}} = \frac{\sin^{-1}(b) - \sin^{-1}a}{b - a} \rightarrow \text{①}$$

$$a < c < b$$

$$a^2 < c^2 < b^2$$

$$a^2 - 1 < c^2 - 1 < b^2 - 1$$

$$1 - a^2 > 1 - c^2 > 1 - b^2$$

$$\sqrt{1-a^2} > \sqrt{1-c^2} > \sqrt{1-b^2}$$

$$\frac{1}{\sqrt{1-a^2}} < \frac{1}{\sqrt{1-c^2}} < \frac{1}{\sqrt{1-b^2}}$$

$$\frac{1}{\sqrt{1-a^2}} < \frac{\sin^{-1}b - \sin^{-1}a}{b-a} < \frac{1}{\sqrt{1-b^2}}$$

$$\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1}b - \sin^{-1}a < \frac{b-a}{\sqrt{1-b^2}}$$

$$b = \frac{3}{5}, a = \frac{1}{2}$$

$$\frac{\frac{3}{5} - \frac{1}{2}}{\sqrt{1 - \frac{1}{4}}} < \sin^{-1} \frac{3}{5} - \sin^{-1} \frac{1}{2} < \frac{\frac{3}{5} - \frac{1}{2}}{\sqrt{1 - \frac{9}{25}}}$$

$$\frac{1}{10} \times \frac{2}{\sqrt{3}} < \sin^{-1} \frac{3}{5} - \frac{\pi}{6} < \frac{1}{10} \times \frac{1}{\sqrt{\frac{16}{25}}}$$

$$\frac{1}{5\sqrt{3}} < \sin^{-1} \frac{3}{5} - \frac{\pi}{6} < \frac{1}{10} \times \frac{5}{4}$$

$$\frac{1}{5\sqrt{3}} < \sin^{-1} \frac{3}{5} - \frac{\pi}{6} < \frac{1}{8} + \pi$$

$$\frac{1}{5\sqrt{3}} + \frac{\pi}{6} < \sin^{-1} \frac{3}{5} < \frac{1}{8} + \frac{\pi}{6}$$

Q Find the region in which $f(x) = 1 - 4x^2 - x^2$ is increasing and decreasing

sol given $f(x) = 1 - 4x - x^2$

$$f'(x) = -4 - 2x$$

$$-2(2+x)$$

$$\text{at } x = -2 \quad f'(x) = 0$$

$$\text{for } x > -2, f'(x) < 0$$

$f(x)$ is decreasing in interval $(-2, \infty)$

$$\text{if } x < -2 \quad f'(x) \geq 0$$

$f(x)$ is increasing in interval $(-\infty, -2)$

Q) prove using mean value theorem $|\sin u - \sin v| \leq |u - v|$

sol) case i: If $u = v$ then nothing to prove

case ii: if $u < v$

let $f(x) = \sin x$ defined in interval (u, v)

a) $f(x)$ is continuous

b) $f'(x) = \cos x$

$f(x)$ is differentiable in (u, v)

$$\text{By LMVT } f'(c) = \frac{f(v) - f(u)}{v - u}$$

$$\cos c = \frac{\sin v - \sin u}{v - u}$$

we know that range of $\cos \theta$ is $[-1, 1]$

$$-1 \leq \cos c \leq 1$$

$$-1 \leq \frac{\sin v - \sin u}{v - u} \leq 1$$

$$\left| \frac{\sin v - \sin u}{v - u} \right| \leq 1$$

$$|\sin v - \sin u| \leq |v - u|$$

case iii) $u > v$

let $f(x) = \sin x$ defined in interval (v, u)

a) $f(x)$ is continuous

b) $f'(x) = \cos x$

$f(x)$ is differentiable in (v, u)

$$\text{By LMVT } f'(c) = \frac{f(u) - f(v)}{u - v}$$

$$\cos c = \frac{\sin u - \sin v}{u - v}$$

w.k.t range of $\cos \theta$ is $[-1, 1]$

$$-1 \leq \cos c \leq 1$$

$$-1 \leq \frac{\sin u - \sin v}{u - v} \leq 1$$

$$\left| \frac{\sin u - \sin v}{u - v} \right| \leq 1$$

$$|\sin u - \sin v| \leq |u - v|$$

a) using mean value theorem prove that $0 < a < b \Rightarrow 1 - \frac{a}{b} < \log \frac{b}{a} < \frac{b}{a} - 1$

also s.t. $\frac{1}{6} < \log \frac{6}{5} < \frac{1}{5}$

sol) let $f(x) = \log x$ defined in $[a, b]$

a) $f(x)$ is continuous in $[a, b]$

b) $f'(x) = \frac{1}{x}$

$f(x)$ is differentiable in (a, b)

By LMVT $\exists c \in (a, b) \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$

$$\frac{1}{c} = \frac{\log b - \log a}{b - a} \Rightarrow \frac{1}{c} = \frac{\log \frac{b}{a}}{b - a}$$

$c \in (a, b)$

$$a < c < b$$

$$\frac{1}{a} > \frac{1}{c} > \frac{1}{b}$$

$$\frac{1}{a} > \frac{\log \frac{b}{a}}{b - a} > \frac{1}{b}$$

$$\frac{b - a}{a} > \log \frac{b}{a} > \frac{b - a}{b}$$

$$\frac{b}{a} - 1 > \log \frac{b}{a} > 1 - \frac{a}{b}$$

$$1 - \frac{a}{b} < \log \frac{b}{a} < \frac{b}{a} - 1$$

Put $a = 5, b = 6$

$$1 - \frac{5}{6} < \log \frac{6}{5} < \frac{6}{5} - 1$$

$$\frac{1}{6} < \log \frac{6}{5} < \frac{1}{5}$$

a) Prove that $x > 0 \quad x - \frac{x^2}{2} < \log(1+x) < x - \frac{x^2}{2(1+x)}$

sol) let $f(x) = x - \frac{x^2}{2} - \log(1+x)$

$$f'(x) = 1 - x \Rightarrow \frac{1}{1+x}$$

$$= \frac{1+x-x-x^2-1}{1+x} = \frac{-x^2}{1+x}$$

$$f'(x) < 0 \text{ for } x > 0 \Rightarrow f(x) < 0$$

$f(x)$ is decreasing function

$$x - \frac{x^2}{2} - \log(1+x) < 0$$

$$x - \frac{x^2}{2} < \log(1+x)$$

$$\text{let } g(x) = \log(1+x) - x + \frac{x^2}{2}$$

$$g'(x) = \frac{1}{1+x} - 1 + \left\{ \frac{2(1+x)(2x) - 2^2 \cdot 1}{4(1+x)^2} \right\}$$

$$= \frac{-x}{1+x} + \left\{ \frac{4x + 4x^2 - 2x^2}{4(1+x)^2} \right\}$$

$$= \frac{-x}{1+x} + \frac{4x + 2x^2}{4(1+x)^2}$$

$$= \frac{-x}{1+x} + \frac{2x(x+2)}{4(1+x)^2}$$

$$= \frac{-x}{1+x} + \frac{x(x+2)}{2(1+x)^2}$$

$$= \frac{-2x(1+x) + x(2+x)}{2(1+x)^2}$$

$$= \frac{-2x - 2x^2 + 2x + x^2}{2(1+x)^2} = \frac{-x^2}{2(1+x)^2} < 0$$

As $g'(x) < 0$ for $x > 0$, $g(x)$ is decreasing function
 $g(x) < 0$

$$\log(1+x) < x - \frac{x^2}{2(1+x)}$$

Cauchy's mean value theorem:

Let $f(x), g(x)$ be 2 functions defined in $[a, b]$ &

a) f, g are continuous in $[a, b]$

b) f, g are differentiable in (a, b)

$$\text{Then } \exists c \in (a, b) \Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Q) Find c by using Cauchy's mean value theorem for $f(x) = \sqrt{x}$, $g(x) = \frac{1}{\sqrt{x}}$ in $[a, b]$

Sol) a) f, g are continuous in $[a, b]$

$$b) f'(x) = \frac{1}{2\sqrt{x}}; g'(x) = -\frac{1}{2}x^{-3/2}$$

f, g are differentiable in (a, b)

$$\text{By CMVT } \exists c \in (a, b) \Rightarrow \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\frac{\frac{1}{2\sqrt{c}}}{-\frac{1}{2}c^{-3/2}} = \frac{\sqrt{b} - \sqrt{a}}{\frac{1}{\sqrt{b}} - \frac{1}{\sqrt{a}}}$$

$$\frac{-c^{-1/2}}{c^{-3/2}} = \frac{\sqrt{b} - \sqrt{a}}{\frac{\sqrt{a} - \sqrt{b}}{\sqrt{ab}}}$$

$$-c^{-\frac{1}{2} + \frac{3}{2}} = \left(\frac{\sqrt{b} - \sqrt{a}}{\sqrt{b} - \sqrt{a}} \times \sqrt{ab} \right)$$

$$-c = -\sqrt{ab}$$

$$c = \sqrt{ab} \in (a, b)$$

Q) $f(x) = e^x$, $g(x) = e^{-x}$ (3,7)

sol) (a) f, g are continuous in $[3,7]$

(b) $f'(x) = e^x$, $g'(x) = -e^{-x}$

f, g are differentiable in $(3,7)$

By CMT $\exists c \in (3,7) \Rightarrow \frac{f(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$

$$\frac{e^c}{-e^{-c}} = \frac{e^7 - e^3}{e^7 - e^3}$$

$$-e \cdot e^c = \frac{e^7 - e^3}{\frac{1}{e^7} - \frac{1}{e^3}}$$

$$-e^{2c} = \frac{e^7 - e^3}{\frac{e^3 - e^7}{e^7 e^3}}$$

$$e^{2c} = \left[\frac{e^7 - e^3}{e^7 - e^3} \times e^{7+3} \right]$$

$$e^{2c} = e^{10}$$

$$2c = 10$$

$$\boxed{c=5} \in (3,7)$$

Q) $f(x) = \log x$, $g(x) = x^2$ $[a,b]$ P.T $\frac{\log b - \log a}{b - a} = \frac{a+b}{2c^2}$

sol) (a) f, g are continuous in $[a,b]$

(b) $f'(x) = \frac{1}{x}$, $g'(x) = 2x$

f, g are differentiable in (a,b)

By CMT $\exists c \in (a,b) \Rightarrow \frac{f(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$

$$\frac{\frac{1}{c}}{2c} = \frac{\log b - \log a}{b^2 - a^2}$$

$$\frac{1}{2c^2} = \frac{\log b - \log a}{(a+b)(b-a)}$$

$$\frac{b+a}{2c^2} = \frac{\log b - \log a}{b-a}$$

Q) verify CMVT for $f(x)$ and $f'(x)$ given $f(x) = \log x$ [1, e]
 solⁿ let $f(x) = \log x$ and $g(x) = f'(x) = \frac{1}{x}$

① f, g are continuous

② $f'(x) = \frac{1}{x}$ $g'(x) = -\frac{1}{x^2}$

f, g are differentiable

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

$$\frac{\frac{1}{c}}{-\frac{1}{c^2}} = \frac{\log e - \log 1}{\frac{1}{e} - \frac{1}{1}}$$

$$-c = \frac{1-0}{\frac{1-e}{e}}$$

$$-c = \frac{1}{1-e} \times e$$

$$c = \frac{e}{e-1}$$

Taylor's series:

If f is a function for $[a, b]$ such that

a) $f^{(n-1)}$ is continuous in $[a, b]$

b) $f^{(n-1)}$ derivative is derivable in (a, b) then

$$f(b) = f(a) + \frac{b-a}{1!} f'(a) + \frac{(b-a)^2}{2!} f''(a) + \dots$$

In general Taylor's series is given by

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$$

Maclaurins series:

If the expansion is about the origin i.e. $a=0$ then, the series is given by

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots$$

Q. Obtain the Taylor's series expansion of $\sin x$ in powers of $x = \frac{\pi}{4}$

sol.

we know that Taylor's series is given by

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots \rightarrow (1)$$

$$\text{Here } a = \frac{\pi}{4}$$

Given, $f(x) = \sin x$

$$f\left(\frac{\pi}{4}\right) = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$f'(x) = \cos x \Rightarrow f'\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f''(x) = -\sin x \Rightarrow f''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$f'''(x) = -\cos x \Rightarrow f'''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$f^{(4)}(x) = \sin x \Rightarrow f^{(4)}\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f^{(5)}(x) = \cos x \Rightarrow f^{(5)}\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin x = \frac{1}{\sqrt{2}} + (x - \frac{\pi}{4}) \frac{1}{\sqrt{2}} - \frac{(x - \frac{\pi}{4})^2}{2!} \frac{1}{\sqrt{2}} - \frac{(x - \frac{\pi}{4})^3}{3!} \frac{1}{\sqrt{2}} + \frac{(x - \frac{\pi}{4})^4}{4!} \frac{1}{\sqrt{2}} + \frac{(x - \frac{\pi}{4})^5}{5!} \frac{1}{\sqrt{2}} - \dots$$

Q) $f(x) = \log(\sin x)$ about $x=3$

sol) $f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$

here $a=3$

$f(3) = \log(\sin 3)$

$f'(x) = \frac{1}{\sin x} \cos x = \cot x \Rightarrow f'(3) = \cot 3$

$f''(x) = -\operatorname{cosec}^2 x \Rightarrow f''(3) = -\operatorname{cosec}^2 3$

$f'''(x) = (-2\operatorname{cosec} x)(-\operatorname{cosec} x \cot x) = 2\operatorname{cosec}^2 x \cot x$

$f'''(3) = 2\operatorname{cosec}^2 3 \cot 3$

$\log(\sin x) = \log \sin 3 + \frac{(x-3)}{1!} \cot 3 - \frac{(x-3)^2}{2!} \operatorname{cosec}^2 3 + \frac{(x-3)^3}{3!} 2\operatorname{cosec}^2 3 \cot 3 + \dots$

Q) obtain maclaurins series for $f(x) = \cos x$

sol) we know that maclaurins series is given by

$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

$f(x) = \cos x \Rightarrow f(0) = \cos 0 = 1$

$f'(x) = -\sin x \Rightarrow f'(0) = 0$

$f''(x) = -\cos x \Rightarrow f''(0) = -1$

$f'''(x) = \sin x \Rightarrow f'''(0) = 0$

$f^{(4)}(x) = \cos x \Rightarrow f^{(4)}(0) = 1$

$1 + \frac{x}{1!} (0) + \frac{x^2}{2!} (-1) + \frac{x^3}{3!} (0) + \frac{x^4}{4!} (1) + \dots$

$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

Q) obtain maclaurins series for $f(x) = \sin x$

sol) $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

$f(x) = \sin x \Rightarrow f(0) = \sin 0 = 0$

$f'(x) = \cos x \Rightarrow f'(0) = \cos 0 = 1$

$f''(x) = -\sin x \Rightarrow f''(0) = -\sin 0 = 0$

$f'''(x) = -\cos x \Rightarrow f'''(0) = -\cos 0 = -1$

$\frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

Q) obtain Maclaurin's series of $f(x) = e^x$

sol) $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots$

$$f(x) = e^x \Rightarrow f(0) = e^0 = 1$$

$$f'(x) = e^x \Rightarrow f'(0) = e^0 = 1$$

$$f''(x) = e^x \Rightarrow f''(0) = e^0 = 1$$

$$1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Q) show that $\frac{\sin^{-1} x}{\sqrt{1-x^2}} = x + \frac{x^3}{3!} + \dots$

sol) $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots$

$$f(x) = \frac{\sin^{-1} x}{\sqrt{1-x^2}} \Rightarrow f(0) = \frac{\sin^{-1}(0)}{\sqrt{1-0}} = 0$$

$$\sqrt{1-x^2} f(x) = \sin^{-1} x$$

$$\sqrt{1-x^2} f'(x) + f(x) \frac{1}{2\sqrt{1-x^2}} (0-2x) = \frac{1}{\sqrt{1-x^2}}$$

$$(1-x^2) f'(x) - x f(x) = 1 \rightarrow \textcircled{1}$$

at $x=0$

$$f'(0) - 0 = 1$$

$$f'(0) = 1$$

again differentiating $\textcircled{1}$ w.r.t. x

$$[(1-x^2) f''(x) + f'(x)(0-2x)] - [x f'(x) + f(x) \cdot 1] = 0$$

$$(1-x^2) f''(x) - 3x f'(x) - f(x) = 0 \rightarrow \textcircled{2}$$

at $x=0$

$$(1-0) f''(0) - 0 - f(0) = 0$$

$$f''(0) = 0$$

differentiating $\textcircled{2}$ w.r.t. x

$$[(1-x^2) f'''(x) + (0-2x) f''(x)] - 3[x f''(x) + f'(x) \cdot 1] - f'(x) = 0$$

Put $x=0$

$$f'''(x) - 4f'(0) = 0$$

$$f'''(x) - 4(1) = 0$$

$$f'''(x) = 4$$

Maclaurin's series

$$\frac{\sin^{-1} x}{\sqrt{1-x^2}} = 0 + \frac{x}{1!}(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(4) + \dots$$

$$\frac{\sin^{-1} x}{\sqrt{1-x^2}} = x + \frac{4x^3}{3!} + \dots$$

Q) show that $\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots$ also

prove that $\frac{e^x}{1+e^x} = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \dots$

Sol) $\frac{e^x}{1+e^x} = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \dots$

$$f(x) = \log(1+e^x) \Rightarrow f(0) = \log(1+e^0) = \log 2$$

$$f'(x) = \frac{e^x}{1+e^x} \Rightarrow f'(0) = \frac{1}{1+1} = \frac{1}{2}$$

$$f''(x) = \frac{(1+e^x) \cdot e^x - e^x(0+e^x)}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}$$

$$\boxed{f''(0) = \frac{1}{4}}$$

$$f'''(x) = \frac{(1+e^x)^2 e^x - e^x \cdot 2(1+e^x) \cdot e^x}{(1+e^x)^4}$$

$$= \frac{(1+e^x) [(1+e^x)e^x - 2e^{2x}]}{(1+e^x)^4}$$

$$= \frac{e^x + e^{2x} - 2e^{2x}}{(1+e^x)^3} = \frac{e^x - e^{2x}}{(1+e^x)^3}$$

$$f'''(0) = \frac{e^0 - e^{2(0)}}{(1+e^0)^3} = \frac{1-1}{(1+e^0)^3} = 0$$

$$f''(x) = \frac{(1+e^x)^3 [e^x - 2e^{2x}] - (e^x - e^{2x}) [3(1+e^x)^2 \cdot e^x]}{(1+e^x)^6}$$

$$= \frac{(1+e^x)^2 [(1+e^x)(e^x - 2e^{2x}) - 3(e^x - e^{2x})(e^x)]}{(1+e^x)^6}$$

$$= \frac{(1+e^x)(e^x - 2e^{2x}) - 3e^x(e^x - e^{2x})}{(1+e^x)^4} = \frac{(1+1)(1-2)}{2^4}$$

$$= \frac{2(-1)}{2^4} = -\frac{2}{2^4} = -\frac{1}{8}$$

$$\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{2!} \cdot \frac{1}{4} + \frac{x^3}{3!} \cdot \frac{1}{8} + \frac{x^4}{4!} \left(-\frac{1}{8}\right)$$

$$= \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots$$

$$\frac{e^x}{1+e^x}$$

$$f(x) = \frac{e^x}{1+e^x}$$

$$f(0) = \frac{1}{2}$$

$$f'(x) = \frac{(1+e^x)e^x - e^x(e^x)}{(1+e^x)^2} = \frac{(1+e^x)e^x - e^{2x}}{(1+e^x)^2}$$

$$= \frac{e^x + e^{2x} - e^{2x}}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2} \Rightarrow f'(0) = \frac{e^0}{(1+e^0)^2} = \frac{1}{4}$$

$$f'(0) = \frac{1}{4}$$

$$\begin{aligned}
 f''(x) &= \frac{(1+e^x)^2 e^x - e^x (2(1+e^x)(e^x))}{(1+e^x)^4} \\
 &= \frac{(1+e^x)^2 \cdot e^x - e^x (2(1+e^x)(e^x))}{(1+e^x)^4} \\
 &= \frac{(1+e^x)^2 e^x - 2e^{2x} (1+e^x)}{(1+e^x)^4} \\
 &= \frac{(1+e^x) [(1+e^x) e^x - 2e^{2x}]}{(1+e^x)^4} \\
 &= \frac{(1+e^x) e^x - 2e^{2x}}{(1+e^x)^3} \\
 &= \frac{e^x + e^{2x} - 2e^{2x}}{(1+e^x)^3} = \frac{e^x - e^{2x}}{(1+e^x)^3}
 \end{aligned}$$

$$f''(0) = \frac{e^0 - e^{2(0)}}{(1+e^0)^3} = \frac{1-1}{(1+1)^3} = 0$$

$$f'''(x) = \frac{f''(x) (1+e^x)^3 [e^x - 2e^{2x}] - 3(e^x - e^{2x})(e^x)}{(1+e^x)^4}$$

$$\begin{aligned}
 f'''(0) &= \frac{(1+e^0) (e^0 - 2e^{2(0)}) - 3(e^0 - e^{2(0)})(e^0)}{(1+e^0)^4} \\
 &= \frac{(1+1)(1-2) - 3(1-1)(1)}{(1+1)^4} = \frac{2(-1)}{2^4} = \frac{-2}{2^4} = -\frac{1}{8}
 \end{aligned}$$

$$\begin{aligned}
 \frac{e^x}{1+e^x} &= \frac{1}{2} + \frac{x}{4} + \frac{x^2}{2!} \times 0 + \frac{x^3}{3!} \times \frac{-1}{8} + \dots \\
 &= \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \dots
 \end{aligned}$$

Applications of Integration

- ① Volumes of solids of evaluation
- ② surface area of revolution

Volumes of solid of evaluation: (Cartesian form)

1) In $y=f(x)$ and axis of revolution is x -axis bounded by $x=a$ & $x=b$ then volume $V = \pi \int_a^b y^2 dx$

2) If curve $x=f(y)$ axis of revolution is y -axis bounded by $y=a$ & $y=b$ then $V = \pi \int_a^b x^2 dy$

3) If $y=y_1(x)$, $y=y_2(x)$ axis of revolution is x axis and the ordinates are $x=a$, $x=b$ then

$$V = \pi \int_a^b (y_2^2 - y_1^2) dx$$

If the curves are $x=x_1(y)$, $x=x_2(y)$ axis of revolution is y axis and the ordinates are $y=a$, $y=b$ then

$$V = \pi \int_a^b (x_2^2 - x_1^2) dy$$

Q) Find the volume of the sphere of radius 'a'
 A sphere is obtained by rotating circles about x -axis are
 know that equation of circle is $x^2 + y^2 = a^2$

$$V = \pi \int_a^b y^2 dx = \pi \int_a^b (a^2 - x^2) dx$$

limits $\rightarrow -a$ to a

$$V = \pi \int_{-a}^a (a^2 - x^2) dx$$

$$= 2\pi \int_0^a (a^2 - x^2) dx$$

$$= 2\pi \left[\int_0^a a^2 dx - \int_0^a x^2 dx \right]$$

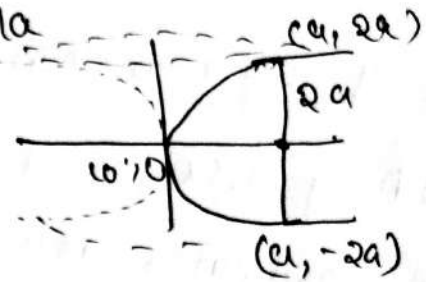
$$= 2\pi \left[a^2 [x]_0^a - \left[\frac{x^3}{3} \right]_0^a \right]$$

$$\begin{aligned}
 & 2\pi \left[a^2 [a] - \left[\frac{a^3}{3} \right] \right] \\
 &= 2\pi \left[a^3 - \frac{a^3}{3} \right] = 2\pi \left[\frac{3a^3 - a^3}{3} \right] = 2\pi \left[\frac{2a^3}{3} \right] \\
 &= \frac{4a^3}{3} \pi = \frac{4}{3} \pi a^3
 \end{aligned}$$

Q) Find the volume of ring shape solid formed by the revolution of parabola $y^2 = 4ax$ cut off by the latus rectum

sol) Given that $y^2 = 4ax$ is a parabola symmetrical about the x-axis

Here the volume $V = \pi \int_a^b x^2 dy$



$$y^2 = 4ax$$

$$x = \frac{y^2}{4a}$$

$$x^2 = \frac{y^4}{16a^2}$$

limits: $y \rightarrow -2a$ to $2a$

$$V = \pi \int_{-2a}^{2a} \frac{y^4}{16a^2} dy = \frac{\pi}{16} \int_{-2a}^{2a} \frac{y^4}{a^2} dy$$

$$= \frac{2\pi}{16a^2} \int_0^{2a} y^4 dy = \frac{\pi}{8a^2} \left[\frac{y^5}{5} \right]_0^{2a}$$

$$= \frac{\pi}{40a^2} [(2a)^5 - 0] = \frac{\pi \times 32a^5}{40a^2}$$

$$= \frac{4\pi}{5} a^3$$

Q) Find the volume of solid generated by revolving ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ about major axis } (0 < b < a)$$

sol) Given volume of revolution is about major axis i.e x-axis

$$V = \pi \int_a^b y^2 dx$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = b^2 \left[1 - \frac{x^2}{a^2} \right]$$

limits: $x \rightarrow -a \text{ to } a$

$$V = \pi \int_{-a}^a b^2 \left[1 - \frac{x^2}{a^2} \right] dx$$

$$V = b^2 \pi \int_{-a}^a \left(1 - \frac{x^2}{a^2} \right) dx$$

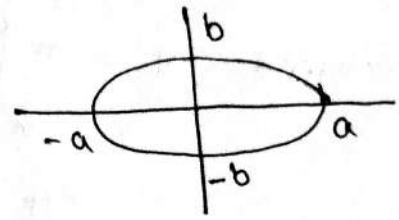
$$V = \frac{b^2 \pi}{a^2} \int_{-a}^a (a^2 - x^2) dx$$

$$= \frac{b^2 \pi}{a^2} \left[\int_{-a}^a a^2 dx - \int_{-a}^a x^2 dx \right]$$

$$= \frac{b^2 \pi}{a^2} \left[(a^2 x)_{-a}^a - \left(\frac{x^3}{3} \right)_{-a}^a \right]$$

$$= \pi \frac{b^2}{a^2} \left[a^2 (a+a) - \frac{1}{3} [a^3 + a^3] \right]$$

$$= \pi \frac{b^2}{a^2} \left[2a^3 - \frac{2}{3} a^3 \right] = \frac{4}{3} \pi b^2 a$$



Q) Find the volume formed by revolution of loop of the curve $y^2(a+x) = x^2(3a-x)$ about the x-axis.

sol) volume $V = \pi \int_a^b y^2 dx$

$$= \pi \int_a^b \frac{x^2(3a-x)}{a+x} dx$$

limits: $y=0 \Rightarrow x^2(3a-x)=0$
 $x=0, x=3a$

$$V = \pi \int_0^{3a} \frac{x^2(3a-x)}{(a+x)} dx$$

$$V = \pi \int_0^{3a} \frac{3ax^2 - x^3}{a+x} dx$$

$$\pi \int_0^{3a} \left(-x^2 + 4ax - 4a^2 + \frac{4a^3}{a+x} \right) dx$$

$$= \pi \left[-\frac{x^3}{3} + 4a \frac{x^2}{2} - 4a^2 \cdot x + 4a^3 \log(a+x) \right]_0^{3a}$$

$$= \pi \left[-\frac{1}{3}(27a^3) + 2a(9a^2) - 4a^2(3a) + 4a^3 \log 4a \right] - \pi [0 + 0 + 4a^3 \log a]$$

$$= \pi \left[-9a^3 + 18a^3 - 12a^3 + 4a^3 \log 4a - 4a^3 \log a \right]$$

$$= \pi \left[-3a^3 + 4a^3 \log 4 \right] = a^3 \pi [4 \log 4 - 3]$$

Q) A sphere of radius 'a' units is divided into two parts by a plane distance $\frac{a}{2}$ from the centre show that the ratio of the volume is 5:27

sol) we know that a sphere is formed by revolving the circle about the x-axis. The equation of circle is $x^2 + y^2 = a^2$. Given sphere is divided into 2 parts by a plane

for the 1st part

$$x \rightarrow -a \text{ to } a/2$$

$$V = \pi \int_{-a}^{a/2} (a^2 - x^2) dx = \pi \left\{ a^2 x - \frac{x^3}{3} \right\}_{-a}^{a/2}$$

$$= \pi a^2 \left[\frac{a}{2} - (-a) \right] - \frac{\pi}{3} \left[\left(\frac{a}{2} \right)^3 - (-a)^3 \right]$$

$$= \pi a^2 \left[\frac{3a}{2} \right] - \frac{\pi}{3} \left[\frac{a^3}{8} + a^3 \right] = \pi a^2 \left[\frac{3a}{2} \right] - \frac{\pi}{3} \left[\frac{9a^3}{8} \right]$$

$$\pi a^2 \left[\frac{3a}{2} - \frac{3a^3}{8} \right] = \frac{3\pi a^3}{2} - \frac{3\pi a^3}{8} = 3\pi a^3 \left[\frac{1}{2} - \frac{1}{8} \right]$$

$$= 3\pi a^3 \left[\frac{4-1}{8} \right]$$

$$= 3\pi a^3 \left[\frac{3}{8} \right]$$

$$= \pi a^3 \left[\frac{9}{8} \right]$$

for part 2:

$$x \rightarrow \frac{a}{2} \text{ to } a$$

$$V = \pi \int_{a/2}^a y^2 dx = \pi \int_{a/2}^a (a^2 - x^2) dx = \pi \left\{ [a^2 x]_{a/2}^a - \left[\frac{x^3}{3} \right]_{a/2}^a \right\}$$

$$= \pi \left\{ a^2 \left[a - \frac{a}{2} \right] - \frac{1}{3} \left[a^3 - \left(\frac{a}{2} \right)^3 \right] \right\}$$

$$= \pi \left\{ a^2 \left[\frac{a}{2} \right] - \frac{1}{3} \left[a^3 - \frac{a^3}{8} \right] \right\} = \pi \left\{ \frac{a^3}{2} - \frac{1}{3} \times \frac{7a^3}{8} \right\}$$

$$= \pi \left[\frac{a^3}{2} - \frac{7a^3}{24} \right] = \pi \left[\frac{12a^3 - 7a^3}{24} \right] = \pi \left[\frac{5a^3}{24} \right]$$

$$\frac{9}{8} \pi a^3 : \frac{5}{24} \pi a^3$$

$$\frac{3}{3} \times \frac{9}{8} : \frac{5}{24}$$

$$\frac{27}{24} : \frac{5}{24}$$

$$27 : 5$$

Surface area of revolution:

1) If the revolution is about x axis and the equation of curve is $y=f(x)$ then surface area $S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

2) If the revolution is about y axis and the equation of curve is $x=f(y)$ then surface area $S = 2\pi \int_a^b x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

Q) Find the surface area of the sphere generated by the circle $x^2 + y^2 = 16$ about it's diameter

sol) let the revolution is about x-axis then surface area

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$x^2 + y^2 = 16$$

$$w.r.t. x$$

$$2x + 2y \left(\frac{dy}{dx}\right) = 0$$

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$\begin{aligned} S &= 2\pi \int_a^b y \sqrt{1 + \frac{x^2}{y^2}} dx = 2\pi \int_{-4}^4 y \frac{\sqrt{y^2 + x^2}}{y} dx \\ &= 2\pi \int_{-4}^4 \sqrt{16} dx = 2\pi \int_{-4}^4 4 dx = 8\pi \int_{-4}^4 dx = 8\pi [x]_{-4}^4 \\ &= 8\pi [4 + 4] \\ &= 8\pi (8) \\ &= 64\pi \end{aligned}$$

Q) Find the area of surface of revolution generated by revolving one arc of the curve $y = \sin x$ about the x axis,

sol) Given $y = \sin x$

$$\frac{dy}{dx} = \cos x$$

$$S = 2\pi \int_0^{\pi} y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_0^{\pi} \sin x \sqrt{1 + (\cos x)^2}$$

Put $\cos x = t$
 $-\sin x dx = dt$

limits

$$x=0 \Rightarrow \cos x = t \Rightarrow t = 1$$

$$x=\pi \Rightarrow \cos \pi = t \Rightarrow t = -1$$

$$S = -2\pi \int_1^{-1} \sqrt{1+t^2} dt = 2\pi \int_{-1}^1 \sqrt{1+t^2} dt$$

$$= 4\pi \left[\frac{t}{2} \sqrt{1+t^2} + \frac{1}{2} \sinh^{-1} t \right]_0^1$$

$$= 4\pi \left\{ \left(\frac{1}{2} \sqrt{2} + \frac{1}{2} \sinh^{-1}(1) \right) \right\} - \{0-0\}$$

$$= 4\pi \left[\frac{1}{\sqrt{2}} + \frac{1}{2} \sinh^{-1}(1) \right]$$

Q) find the surface area generated by revolution of an arc

$y = c \cosh \frac{x}{c}$ about the x axis from $x=0$ to c

Sol) $y = c \cosh \frac{x}{c}$; $S = 2\pi \int_0^c y \cdot \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

$$\frac{dy}{dx} = \sinh \frac{x}{c} \cdot \frac{1}{c}$$

$$\frac{dy}{dx} = \sinh \frac{x}{c}$$

$$S = 2\pi \int_0^c \cosh \frac{x}{c} \sqrt{1 + \sinh^2 \frac{x}{c}} dx$$

$$= 2\pi c \int_0^c \cosh^2 \frac{x}{c} dx$$

$$= 2\pi c^2 \int_0^1 \cosh^2 \theta d\theta$$

$$= 2\pi c^2 \int_0^1 \frac{1 + \cosh 2\theta}{2} d\theta = \pi c^2 \left[\theta + \frac{\sinh 2\theta}{2} \right]_0^1$$

$$= \pi c^2 \left\{ \left[1 + \frac{\sinh 2\theta}{2} \right] - (0-0) \right\}$$

$$= \pi c^2 \left[1 + \frac{\sinh 2\theta}{2} \right]$$

Q) Find the surface area where the loop of the curve $3ay^2 = x(x-a)^2$ is revolved about x-axis.

Sol) $3ay^2 = x(x-a)^2$

$$6ay \frac{dy}{dx} = x(x^2 + a^2 - 2ax)$$

$$6ay \frac{dy}{dx} = x^3 + a^2x - 2ax^2$$

$$= 3x^2 + a^2 - 4ax$$

$$\frac{dy}{dx} = \left[\frac{3x^2 + a^2 - 4ax}{6ay} \right]$$

$$S = 2\pi \int_0^a y \sqrt{1 + \left(\frac{3x^2 + a^2 - 4ax}{6ay} \right)^2} dx$$

$$S = 2\pi \int_0^a y \frac{\sqrt{36a^2y^2 + (3x^2 + a^2 - 4ax)^2}}{6ay} dx = \pi \int_0^a \frac{\sqrt{36a^2y^2 + (3x-a)^2(x-a)^2}}{3a} dx$$

$$6ay \cdot \frac{dy}{dx} = 2x(x-a) + (x-a)^2$$

$$6ay \cdot \frac{dy}{dx} = 2x^2 - 2ax + (x-a)^2$$

$$y=0 \Rightarrow 0 = x(x-a)^2$$

$$x=0; (x-a)=0$$

$$x=a$$

$$x \rightarrow 0 \text{ to } a$$

$$S = 2\pi \int_0^a y \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

$$\begin{aligned}
 &= \frac{\pi}{3a} \int_0^a \left[36a^2 \cdot \frac{x(x-a)^2}{3a} + (3x-a)^2 (x-a)^2 \right]^{1/2} dx \\
 &= \frac{\pi}{3a} \int_0^a \left(12ax (x-a)^2 + (3x-a)^2 (x-a)^2 \right)^{1/2} dx \\
 &= \frac{\pi}{3a} \int_0^a ((x-a)^2)^{1/2} [12ax + (3x-a)^2]^{1/2} dx \\
 &= \frac{\pi}{3a} \int_0^a ((x-a)^2)^{1/2} [9x^2 + a^2 + 6xa]^{1/2} dx \\
 &= \frac{\pi}{3a} \int_0^a \{ (x-a) (3x+a) \} dx \\
 &= \frac{\pi}{3a} \int_0^a 3x^2 - 2ax - a^2 dx \\
 &= \frac{\pi}{3a} \left[3 \frac{x^3}{3} - 2a \frac{x^2}{2} - a^2 x \right]_0^a \\
 &= \frac{\pi}{3a} \left[x^3 - x^2 a - x a^2 \right]_0^a \\
 &= \frac{\pi}{3a} \{ (a^3 - a^2 \cdot a - a \cdot a^2) - (0-0) \} \\
 &= \frac{\pi}{3a} [a^3 - a^3 - a^3] = \frac{-a^3 \pi}{3a} = \frac{-\pi a^2}{3} = \frac{\pi a^2}{3}
 \end{aligned}$$

Beta and Gamma Functions

Beta function: The definite integral $\int_0^1 x^{m-1} (1-x)^{n-1} dx$ is called Beta function denoted by $B(m, n)$ or $\beta(m, n)$ where $m > 0, n > 0$, also known as Eulerian Integral of first kind.

Properties:

① $B(m, n) = B(n, m)$

sol) $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$
 put $1-x=t \Rightarrow -dx = dt$

limits: If $x=0 \Rightarrow t=1$

If $x=1 \Rightarrow t=0$

$$\begin{aligned} B(m, n) &= \int_1^0 (1-t)^{m-1} t^{n-1} (-dt) \\ &= - \int_1^0 t^{n-1} (1-t)^{m-1} dt \\ &= \int_0^1 t^{n-1} (1-t)^{m-1} dt \\ &= B(n, m) \end{aligned}$$

② $B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

sol) By def $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \rightarrow \text{①}$

put $x = \sin^2 \theta$

$dx = 2 \sin \theta \cos \theta d\theta$

$dx = \sin 2\theta d\theta$

limits: If $x=0 \Rightarrow \sin \theta = 0$
 $\theta = 0$

If $x=1 \Rightarrow \sin \theta = 1$

$\theta = \pi/2$

$$\begin{aligned} \textcircled{1} \Rightarrow B(m, n) &= \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} \sin 2\theta \, d\theta \\ &= \int_0^{\pi/2} \sin^{2m-2} \theta \cdot \cos^{2n-2} \theta \cdot 2 \sin \theta \cos \theta \, d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta \, d\theta \end{aligned}$$

$$\textcircled{3} \quad B(m, n) = B(m+1, n) + B(m, n+1)$$

sol) considers

$$B(m+1, n) + B(m, n+1)$$

$$\begin{aligned} &= \int_0^1 x^{(m+1)-1} (1-x)^{n-1} \, dx + \int_0^1 x^{m-1} (1-x)^{n+1-1} \, dx \\ &= \int_0^1 [x^m (1-x)^{n-1} + x^{m-1} (1-x)^n] \, dx \\ &= \int_0^1 x^{m-1} (1-x)^{n-1} \{x + (1-x)\} \, dx \\ &= \int_0^1 x^{m-1} (1-x)^{n-1} \, dx = B(m, n) \end{aligned}$$

$$\textcircled{4} \quad \text{If } m, n \text{ are +ve integers, then } B(m, n) = \frac{(m-1)! (n-1)!}{(m+n-1)!}$$

Other forms of Beta function:

$$\begin{aligned} \text{Form 1: } B(m, n) &= \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} \, dx \\ &= \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} \, dx \end{aligned}$$

1) consider

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \rightarrow (1)$$

Put $x = \frac{1}{1+y} \Rightarrow 1+y = \frac{1}{x}$
 $dx = \frac{-1}{(1+y)^2} dy \quad y = \frac{1}{x} - 1$

Limits: $x=0 \Rightarrow y = \infty$

$x=1 \Rightarrow y = 1-1 = 0$

$$(1) \Rightarrow B(m, n) = \int_{\infty}^0 \left[\frac{1}{1+y} \right]^{m-1} \left[1 - \frac{1}{1+y} \right]^{n-1} - \frac{1}{(1+y)^2} dy$$

$$\Rightarrow B(m, n) = \int_{\infty}^0 - \frac{1}{(1+y)^{m-1+2}} \left[\frac{1+y-1}{1+y} \right]^{n-1} dy$$

$$= \int_0^{\infty} \frac{1}{(1+y)^{m+1}} \cdot \frac{y^{n-1}}{(1+y)^{n-1}} dy$$

$$B(m, n) = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n-1+1}} dy = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy \rightarrow (2)$$

$$B(m, n) = B(n, m)$$

$$(2) \Rightarrow B(m, n) = \int_0^{\infty} \frac{y^{m-1}}{(1+y)^{m+n}} dy$$

FORM II: $B(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$

Sol: By FORM I:

$$B(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx + \int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx \rightarrow (1)$$

consider $\int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx \rightarrow (2)$

Put $x = \frac{1}{y} \Rightarrow dx = -\frac{1}{y^2} dy$

limits $x=1 \Rightarrow y=1$

$x=\infty \Rightarrow y=\frac{1}{x}=0$

$$(2) \Rightarrow \int_1^0 \left[\frac{1}{y}\right]^{m-1} \frac{1}{\left(1+\frac{1}{y}\right)^{m+n}} \because -\frac{1}{y^2} dy$$

$$= - \int_1^0 \frac{1}{y^{m-1}} \frac{y^{m+n}}{(y+1)^{m+n}} \cdot \frac{1}{y^2} dy$$

$$= \int_0^1 \frac{-y^{m+n-2-m+1}}{(1+y)^{m+n}} dy$$

$$= \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}} dy = \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$(1) \Rightarrow B(m, n) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx + \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$= \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$$

Form III: s.t $B(m, n) = a^m b^n \int_0^{\infty} \frac{x^{m-1}}{(ax+b)^{m+n}} dx$

sol) consider

$$a^m b^n \int_0^{\infty} \frac{x^{m-1}}{(ax+b)^{m+n}} dx$$

$$= \frac{a^m b^n}{b^{m+n}} \int_0^{\infty} \frac{x^{m-1}}{\left[\frac{ax}{b} + 1\right]^{m+n}} dx \rightarrow \textcircled{1}$$

put $\frac{ax}{b} = y \Rightarrow \frac{a}{b} dx = dy$

$$dx = \frac{b}{a} dy$$

limits: $x=0 \Rightarrow y=0$

$x=\infty \Rightarrow y=\infty$

$$\textcircled{1} \Rightarrow \frac{a^m}{b^m} \int_0^{\infty} \frac{\left[\frac{by}{a}\right]^{m-1}}{[y+1]^{m+n}} \cdot \frac{b}{a} dy$$

$$\frac{a^m}{b^m} \int_0^{\infty} \frac{b^{m-1}}{a^{m-1}} \cdot \frac{y^{m-1}}{(1+y)^{m+n}} \cdot \frac{b}{a} dy$$

$$= \frac{a^{m-n+1-1}}{b^{m-n+1-1}} \int_0^{\infty} \frac{y^{m-1}}{(1+y)^{m+n}} dy$$

$$= \int_0^{\infty} \frac{y^{m-1}}{(1+y)^{m+n}} dy$$

$= B(m, n)$ [by form I]

Form 1 v: $\int_0^1 \frac{x^{m-1} (1-x)^{n-1}}{(x+a)^{m+n}} dx = \frac{B(m,n)}{x^n (1+a)^m}$

sol) By def $B(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \rightarrow \textcircled{1}$

Put $x = \frac{(1+a)t}{t+a} = (1+a) \frac{t}{t+a}$

$$dx = (1+a) \left\{ \frac{(t+a) \cdot dt - t \cdot dt}{(t+a)^2} \right\}$$

$$= \frac{a(1+a)}{(t+a)^2} dt$$

limits: $\text{L.L } x=0 \Rightarrow 0 = \frac{(1+a)t}{t+a}$

$$(1+a)t = 0 \Rightarrow t = 0$$

U.L : $x=1 \Rightarrow 1 = \frac{(1+a)t}{t+a}$

$$t+a = t+at$$

$$a = at$$

$$t=1$$

$$\textcircled{1} B(m,n) = \int_0^1 \left[\frac{(1+a)t}{t+a} \right]^{m-1} \left[1 - \frac{(1+a)t}{t+a} \right]^{n-1} \times \frac{a(1+a)}{(t+a)^2} dt$$

$$= \int_0^1 (1+a)^{m-1} \frac{t^{m-1}}{(t+a)^{m-1}} \cdot \frac{(a-at)^{n-1}}{(t+a)^{n-1}} \cdot \frac{a(1+a)}{(t+a)^2} dt$$

$$= \int_0^1 (1+a)^{m-1+1} \frac{t^{m-1}}{(t+a)^{m-1+n-1+2}} \cdot a^{n-1} (1-t)^{n-1} \cdot a dt$$

$$= \int_0^1 (1+a)^m \frac{t^{m-1}}{(t+a)^{m+n}} a^n (1-t)^{n-1} dt$$

$$B(m, n) = a^n (1+a)^m \int_0^1 \frac{t^{m-1} (1-t)^{n-1}}{(1-ta)^{m+n}} dt$$

Form V:

$$S = \int_a^b (x-b)^{m-1} (a-x)^{n-1} dx = (a-b)^{m+n-1} B(m, n) \quad m > 0; n > 0$$

$$\int_a^b (x-a)^{m-1} (b-x)^{n-1} dx = (b-a)^{m+n-1} B(m, n)$$

PROOF:

$$\text{By def } \int_0^1 x^{m-1} (1-x)^{n-1} dx \rightarrow 0 = B(m, n)$$

$$\text{Put } x = \frac{t-a}{b-a}$$

$$dx = \frac{dt}{b-a}$$

$$\text{limits: } x=0 \Rightarrow t=a$$

$$x=1 \Rightarrow t=b$$

$$\textcircled{1} \Rightarrow B(m, n) = \int_a^b \left[\frac{t-a}{b-a} \right]^{m-1} \cdot \left[1 - \frac{t-a}{b-a} \right]^{n-1} \frac{dt}{b-a}$$

$$= \int_a^b \frac{(t-a)^{m-1}}{(b-a)^{m-1}} \cdot \frac{(b-t)^{n-1}}{(b-a)^{n-1}} \cdot \frac{dt}{b-a}$$

$$= \int_a^b \frac{(t-a)^{m-1} (b-t)^{n-1}}{(b-a)^{m+n}} dt$$

$$= (b-a)^{m+n} B(m, n) = \int_a^b (t-a)^{m-1} (b-t)^{n-1} dt$$

Q. S.T $\int_0^{\pi/2} \sin^m \theta \cdot \cos^n \theta d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$

sol) By property ②

$$B(p, q) = 2 \int_0^{\pi/2} \sin^{2p-1} \theta \cos^{2q-1} \theta d\theta \rightarrow \text{①}$$

comparing given eq with ①

$$2p-1 = m \Rightarrow p = \frac{m+1}{2}$$

$$2q-1 = n \Rightarrow q = \frac{n+1}{2}$$

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$$

Q. Express in Beta function

a) $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$

sol) By def $\int_0^1 x^{m-1} (1-x)^{n-1} dx = B(m, n)$

$$\int_0^3 [9-x^2]^{-1/2} dx$$

$$= \int_0^3 9^{-1/2} \left[1 - \left(\frac{x}{3}\right)^2\right]^{-1/2} dx \rightarrow \text{①}$$

put $\left(\frac{x}{3}\right)^2 = t \Rightarrow \frac{x}{3} = \sqrt{t}$

$$x = 3\sqrt{t}$$

$$dx = 3 \frac{1}{2\sqrt{t}} dt$$

Limits:

L.L $x=0 \Rightarrow t=0$

U.L $x=3 \Rightarrow 3=3\sqrt{t} \Rightarrow \sqrt{t}=1$
 $t=1$

$$\textcircled{1} \Rightarrow \int_0^1 \frac{1}{3} [1-t]^{1/2} \frac{3}{2} \frac{1}{\sqrt{t}} dt$$

$$= \frac{1}{2} \int_0^1 t^{-1/2} (1-t)^{1/2} dt$$

Here $m-1 = -\frac{1}{2}$ $n-1 = -\frac{1}{2}$

$$m = \frac{1}{2} \quad ; \quad n = \frac{1}{2}$$

$$= \frac{1}{2} B\left[\frac{1}{2}, \frac{1}{2}\right]$$

3) show that $B(m+2, n-2) = \frac{m(m+1)}{(n-1)(n-2)} B(m, n)$

Sol $B(m+2, n-2) = \int_0^1 x^{(m+2)-1} (1-x)^{n-2-1} dx$

$$= \int_0^1 x^{m+1} \cdot (1-x)^{n-3} dx$$

$$= x^{m+1} \frac{(1-x)^{n-2}}{n-2} (-1) \Big|_0^1 - \int_0^1 (m+1) x^m \frac{(1-x)^{n-2} (-1)}{(n-2)} dx$$

$$= \underbrace{[0-0]}_{\text{O.L.L.L.}} + \frac{(m+1)}{n-2} \int_0^1 x^m (1-x)^{n-2} dx \rightarrow \textcircled{1}$$

$$B(m+2, n-2) = \frac{m+1}{n-2} \int_0^1 x^m (1-x)^{n-2} dx$$

$$= \frac{m+1}{n-2} \left\{ x^n \frac{(1-x)^{n-1}}{n-1} (-1) \Big|_0^1 - \int_0^1 m x^{m-1} \frac{(1-x)^{n-1}}{n-1} (-1) dx \right\}$$

$$= \frac{m+1}{n-2} \left\{ 0 + \frac{m}{n-1} \int_0^1 x^{m-1} (1-x)^{n-1} dx \right\}$$

$$B[m+2, n-2] = \frac{(m+1)m}{(n-2)(n-1)} \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$= \frac{(m+1)m}{(n-2)(n-1)} B(m, n)$$

$$dx = \frac{1}{2n} \left[t^{1/2} \right]^{1-2n} dt$$

$$= \frac{1}{2n} t^{\frac{1}{2n}-1} dt$$

$$(2) \Rightarrow \int_0^{\infty} \frac{t^{2m/2n}}{1+t} \cdot \frac{1}{2n} t^{\frac{1-2n}{2n}} dt = \frac{\pi}{2n} \operatorname{cosec} s\pi$$

$$\int_0^{\infty} \frac{t^{\frac{2m-2n+1}{2n}}}{1+t} dt = \pi \operatorname{cosec} s\pi$$

$$\int_0^{\infty} \frac{x^{s-1}}{1+x} dt = \pi \operatorname{cosec} s\pi$$

$$\int_0^{\infty} \frac{x^{n-1}}{1+x} dt = \pi \operatorname{cosec} n\pi$$

$$\Gamma(1-n) \Gamma(n) = \pi \operatorname{cosec} n\pi$$

$$= \frac{\pi}{\sin n\pi}$$

Q) Prove that $\Gamma(1/2) = \sqrt{\pi}$

Sol) $B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$

Put $m=n=1/2$

$$B\left[\frac{1}{2}, \frac{1}{2}\right] = \frac{\Gamma(1/2) \Gamma(1/2)}{\Gamma(1)}$$

$$\{\Gamma(1/2)\}^2 = B\left[\frac{1}{2}, \frac{1}{2}\right] = \int_0^1 x^{\frac{1}{2}-1} (1-x)^{\frac{1}{2}-1} dx$$

$$= \int_0^1 x^{-1/2} (1-x)^{-1/2} dx$$

Put $x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$

limits : L.L $\Rightarrow x=0 \Rightarrow \theta=0$

U.L $\Rightarrow x=1 \Rightarrow \sin \theta = 1 \Rightarrow \theta = \pi/2$

$$\left\{ \sqrt{1/2} \right\}^2 = \int_0^{\pi/2} \sin^1 \theta [1 - \sin^2 \theta]^{-1/2} 2 \sin \theta \cos \theta d\theta$$

$$= \int_0^{\pi/2} \frac{1}{\sin \theta} \cdot \frac{1}{\cos \theta} 2 \sin \theta \cos \theta d\theta$$

$$\left\{ \sqrt{1/2} \right\}^2 = 2 \int_0^{\pi/2} d\theta = 2\theta \Big|_0^{\pi/2} = \pi$$

Q) $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

sol)

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

Put $n = \frac{1}{2}$

$$\Gamma(1/2) = \int_0^{\infty} e^{-x} x^{-1/2} dx$$

Put $x = t^2 \Rightarrow dx = 2t dt$

$$\sqrt{\pi} \int_0^{\infty} e^{-t^2} \cdot t^{-1} 2t dt$$

$$= 2 \int_0^{\infty} e^{-t^2} dt$$

$$\int_0^{\infty} e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$$

Note:

$$1) \int_{-\infty}^0 e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$$2) \int_{-\infty}^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2} + \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$$

Problems:

$$1) \sqrt{11/2}$$

sol) since $\sqrt{n} = (n-1) \sqrt{n-1}$

$$\begin{aligned} \sqrt{11/2} &= \frac{9}{2} \sqrt{9/2} = \frac{9}{2} \cdot \frac{7}{2} \sqrt{7/2} = \frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{1/2} \\ &= \frac{9 \cdot 7 \cdot 5 \cdot 3}{2^5} \sqrt{\pi} \end{aligned}$$

$$2) \sqrt{9}$$

sol) $\sqrt{n} = (n-1)!$

$$\sqrt{9} = 8!$$

$$3) \text{ Find } \sqrt{-7/2}$$

sol) $\sqrt{n+1} = n \sqrt{n}$

$$\sqrt{n} = \frac{\sqrt{n+1}}{n}$$

Put $n = -7/2$

$$\sqrt{-7/2} = \frac{\sqrt{-5/2}}{-7/2} = \frac{-2}{7} \sqrt{-5/2}$$

$$= \frac{-2}{7} \frac{\sqrt{-3/2}}{-5/2} = \frac{-2}{7} \cdot \frac{-2}{5} \sqrt{-3/2}$$

$$= \frac{-2}{7} \cdot \frac{-2}{5} \cdot \frac{-2}{3} \sqrt{1/2}$$

$$= -\frac{2}{7} \cdot -\frac{2}{5} \cdot -\frac{2}{3} \cdot -\frac{2}{1} \sqrt{1/2}$$

$$= \frac{(-2)^4}{7 \cdot 5 \cdot 3} \sqrt{\pi}$$

4) solve $\sqrt{-11/2}$

sol

$$\sqrt{-11/2} = -\frac{2}{11} \cdot -\frac{2}{9} \cdot -\frac{2}{7} \cdot -\frac{2}{5} \cdot -\frac{2}{3} \cdot -\frac{2}{1} \sqrt{1/2}$$

$$= \frac{(-2)^6}{11 \cdot 9 \cdot 7 \cdot 5 \cdot 3} \sqrt{\pi}$$